

gemstone-rs BridgeRoot and Object Mapping

A first MagLev-style bridge-root mapping layer over OOPs



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BridgeRoot and Object Mapping

`gemstone-rs` now has a first object-mapping layer over the lower-level `Oop` API.

This is intentionally smaller than MagLev/GBS object mapping. It gives Rust code a bridge-root dictionary, typed Rust payload mapping, nested read-back, and generated mapping support while keeping direct OOP access available.

What Is Supported

The initial mapping layer supports:

- `Session::bridge_root()`
- `Session::bridge_root_named(name)`
- `BridgeRoot::put`
- `BridgeRoot::put_mapped`
- `BridgeRoot::get_oop`
- `BridgeRoot::get_value`
- `BridgeRoot::get_string`
- `BridgeRoot::get_smallint`
- `BridgeRoot::get_bool`
- `BridgeRoot::get_dictionary`
- `BridgeRoot::get_mapped`
- `BridgeRoot::remove`
- `BridgeRoot::commit`
- `BridgeRoot::commit_with_retry`
- `BridgeRoot::transaction`
- `BridgeDictionary::at_oop`
- `BridgeDictionary::at_value`
- `BridgeDictionary::at_string`
- `BridgeDictionary::at_smallint`
- `BridgeDictionary::at_bool`
- `BridgeDictionary::at_dictionary`
- `BridgeDictionary::at_mapped`
- `BridgeDictionary::at_vec`
- `BridgeKey`
- `BridgeKeyType::String`
- `BridgeKeyType::Symbol`
- `BridgeMapped`
- `#[derive(BridgeMapped)]`
- `BridgeFieldRead`
- `BridgeFieldWrite`

- `BridgeValue::Nil`
- `BridgeValue::Bool`
- `BridgeValue::SmallInt`
- `BridgeValue::String`
- `BridgeValue::Symbol`
- `BridgeValue::Oop`
- `BridgeValue::Dictionary`
- `BridgeValue::KeyedDictionary`
- `BridgeValue::Array`
- session-local OOP identity ids with `Session::identity_for_oop`

The default root is a `GemStone Dictionary` stored in `UserGlobals` under `#GemStoneRsBridgeRoot`.

Rust Example

```
use gemstone_rs::{BridgeValue, Config, Session};

fn main() -> gemstone_rs::Result<()> {
    let mut session = Session::login(Config::from_env())?;

    let payload = BridgeValue::dictionary([
        ("name".to_string(), BridgeValue::from("Tariq")),
        ("amount".to_string(), BridgeValue::from(100_i64)),
        ("currency".to_string(), BridgeValue::from("GBP")),
    ]);

    let mut bridge_root = session.bridge_root()?;
    let payload_oop = bridge_root.put("MyTestDict", payload)?;
    let stored = bridge_root.get_oop("MyTestDict"?);
    assert_eq!(payload_oop, stored);

    bridge_root.commit()?;
    Ok(())
}
```

Run the example:

```
cargo run -p gemstone-rs --example bridge_root_mapping
```

Expected output includes:

```
bridge root: GemStoneRsBridgeRoot
MyTestDict OOP: <number>
```

Typed Rust Struct Mapping

For application code, implement `BridgeMapped` on a plain Rust type. The trait keeps the mapping explicit and reviewable while removing repeated dictionary field reads from application code.

```
use gemstone_rs::{BridgeDictionary, BridgeMapped, BridgeValue, Config, Session};

#[derive(Debug, Eq, PartialEq)]
struct BookingDraft {
    name: String,
    amount: i64,
    currency: String,
}

impl BridgeMapped for BookingDraft {
    fn to_bridge_value(&self) -> BridgeValue {
        BridgeValue::dictionary([
            ("name".to_string(), BridgeValue::from(self.name.clone())),
            ("amount".to_string(), BridgeValue::from(self.amount)),
            ("currency".to_string(), BridgeValue::from(self.currency.clone())),
        ])
    }

    fn from_bridge_dictionary(dictionary: &mut BridgeDictionary<'_>) ->
gemstone_rs::Result<Self> {
        Ok(Self {
            name: dictionary.at_string("name")?,
            amount: dictionary.at_smallint("amount")?,
            currency: dictionary.at_string("currency")?,
        })
    }
}

fn main() -> gemstone_rs::Result<()> {
    let mut session = Session::login(Config::from_env())?;
    let mut bridge_root = session.bridge_root()?;

    let draft = BookingDraft {
        name: "Tariq".to_string(),
        amount: 100,
        currency: "GBP".to_string(),
    };

    bridge_root.put_mapped("MyTestDict", &draft)?;
    let loaded: BookingDraft = bridge_root.get_mapped("MyTestDict")?;
    assert_eq!(loaded, draft);

    bridge_root.commit()?;
    Ok(())
}
```

The checked-in example uses this pattern:

```
cargo run -p gemstone-rs --example bridge_root_mapping
```

Expected output includes:

```
bridge root: GemStoneRsBridgeRoot
MyTestDict OOP: <number>
loaded payload: BookingDraft { name: "Tariq", amount: 100, currency: "GBP" }
```

Derive-Based Mapping

For normal Rust structs, prefer `#[derive(BridgeMapped)]`. The derive writes a `BridgeMapped` implementation that stores fields in a GemStone dictionary and reads them back into Rust.

```
use gemstone_rs::{BridgeMapped, Config, Session};

#[derive(Clone, Debug, Eq, PartialEq, BridgeMapped)]
struct CustomerDraft {
    #[bridge(key_type = "Symbol")]
    name: String,
}

#[derive(Clone, Debug, Eq, PartialEq, BridgeMapped)]
struct BookingDraft {
    #[bridge(key = "amount", key_type = "Symbol")]
    amount: i64,
    customer: CustomerDraft,
    tags: Vec<String>,
}

fn main() -> gemstone_rs::Result<()> {
    let mut session = Session::login(Config::from_env())?;
    let mut bridge_root = session.bridge_root()?;

    let draft = BookingDraft {
        amount: 100,
        customer: CustomerDraft {
            name: "Tariq".to_string(),
        },
        tags: vec!["priority".to_string(), "demo".to_string()],
    };

    bridge_root.put_mapped("DerivedBookingDraft", &draft)?;
    let loaded: BookingDraft = bridge_root.get_mapped("DerivedBookingDraft")?;
    assert_eq!(loaded, draft);

    bridge_root.commit()?;
```

```
Ok(())
}
```

Run the checked-in live example:

```
cargo run -p gemstone-rs --example derive_mapping
```

Expected output includes:

```
derived mapped payload: BookingDraft { amount: 100, customer: CustomerDraft { name:
"Tariq" }, tags: ["priority", "demo"] }
bridge root identity: <number>
```

Nested Read-Back

Nested mapped structs and arrays round-trip through the same bridge dictionary format:

Rust field	GemStone storage	Read-back helper
String	String	BridgeFieldRead / at_string
i64	SmallInteger	BridgeFieldRead / at_smallint
bool	true / false	BridgeFieldRead / at_bool
Oop	raw object reference	BridgeFieldRead / at_oop
another BridgeMapped struct	nested Dictionary	BridgeDictionary::at_mapped
Vec<T>	Array	BridgeDictionary::at_vec

This lets a Rust payload keep normal nested shape:

```
#[derive(Clone, Debug, Eq, PartialEq, BridgeMapped)]
struct BookingDraft {
    customer: CustomerDraft,
    tags: Vec<String>,
}
```

When read-back fails, mapping errors include field context:

```
field amount expected GemStone value type SmallInt, got Oop 1234
```

Key Policy

GemStone dictionaries often use either string keys or symbol keys. The mapping layer makes that explicit:

```
#[derive(Clone, Debug, Eq, PartialEq, BridgeMapped)]
struct BookingDraft {
    #[bridge(key = "amount", key_type = "Symbol")]
    amount: i64,
}
```

Manual code can use the same policy:

```
use gemstone_rs::BridgeKeyType;

bridge_root.put_with_key_type("BookingDraft", BridgeKeyType::Symbol, "stored"?;
let oop = bridge_root.get_oop_with_key_type("BookingDraft", BridgeKeyType::Symbol)?;
```

Use string keys when interoperating with JSON-like payloads and symbol keys when you want Smalltalk-style dictionary access.

Transactions and Identity

`BridgeRoot::transaction` commits on success and aborts on error:

```
bridge_root.transaction(|root| {
    root.put_mapped("BookingDraft", &draft)?;
    let loaded: BookingDraft = root.get_mapped("BookingDraft"?;
    assert_eq!(loaded, draft);
    Ok(())
})?;
```

For conflict-prone commits, use a bounded retry:

```
bridge_root.put_mapped("BookingDraft", &draft)?;
bridge_root.commit_with_retry(2)?;
```

`Session` also tracks a session-local identity id for OOPs it sees:

```
let oop = bridge_root.get_oop("BookingDraft"?;
let identity = bridge_root.identity_id();
println!("bridge root identity={identity}, payload oop={}", oop.raw());
```


That is not transparent object persistence. It is a stable in-session cache key for wrappers, inspectors, and explorer views.

Comparison With MagLev/GBS

The MagLev branch example uses:

```
session bridgeRoot at: #MyTestDict put: payload.  
session commitTransactionOrSignalConflict
```

The closest current `gemstone-rs` shape is:

```
let mut bridge_root = session.bridge_root()?;  
bridge_root.put("MyTestDict", payload)?;  
bridge_root.commit()?;
```

For typed Rust structs:

```
bridge_root.put_mapped("MyTestDict", &draft)?;  
let loaded: BookingDraft = bridge_root.get_mapped("MyTestDict")?;
```

This is still explicit mapping, not transparent persistence. The benefit is that Rust teams can review every field mapping and still keep direct OOP access available when they need it.

Codegen Support

`gemstone-rs` codegen can emit `BridgeMapped` structs from config:

```
mapped = BookingDraft | doc=A typed Rust payload stored under BridgeRoot.  
field = BookingDraft.name | type=String | key=name  
field = BookingDraft.amount | type=SmallInt | key=amount | key_type=Symbol  
field = BookingDraft.currency | type=String | key=currency  
field = BookingDraft.tags | type=Vec<String> | key=tags
```

Generate a mapping config proposal from a live GemStone class:

```
gemstone-rs codegen discover-mapping gemstone-rs.codegen BookingDraft Object
```

Run the live discovery example:

```
cargo run -p gemstone-rs --example codegen_discover_mapping
```

Run the generated mapping example:

```
cargo run -p gemstone-rs --example generated_mapping_app
```

This is the recommended path for repeated mappings because the generated source is checked in and reviewed like any other Rust code.

Explorer and VS Code Support

The local explorer exposes BridgeRoot-oriented endpoints:

```
curl -s http://127.0.0.1:8787/api/bridge/root
curl -s 'http://127.0.0.1:8787/api/bridge/get?key=BookingDraft'
curl -s 'http://127.0.0.1:8787/api/bridge/mapping-config?mapped=BookingDraft'
curl -s 'http://127.0.0.1:8787/api/codegen/discover-mapping?
mapped=BookingDraft&class=Object'
```

The VS Code workbench adds the same object-mapping workflow:

- GemStone RS: Generate Mapping Config
- GemStone RS: Preview BridgeRoot
- GemStone RS: Run Generated Mapping Example

The sidebar also shows these actions under Codegen Config .

Current Boundaries

The mapping layer is explicit, not transparent persistence. It does not yet make every GemStone object look like a native Rust object automatically. The current design is deliberately conservative:

- Oop remains visible and available.
- Session is still non-Send and non-Sync .
- writes happen only through explicit put , put_mapped , or generated code.
- the identity map is session-local and does not replace GemStone identity.

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